

RESCUE VISION

AI-powered drone detection system for missing persons in hostile environments

THE PROBLEM

FINDING PEOPLE IN COMPLEX ENVIRONMENTS IS HARD



Vast Areas

Mountains, forests, and disaster zones cover enormous terrain that's impossible to scan manually.



Human Fatigue

Operators watching hours of drone footage miss critical details — victims hidden behind rocks or vegetation.



Poor Connectivity

Remote locations have limited bandwidth. Streaming raw video from drones is often impractical.

SYSTEM PIPELINE

END-TO-END ON-DEVICE PROCESSING

01



**YOLO
DETECTION**

Fine-tuned on COCO
with degradation
augmentations

02



**DEPTH
ESTIMATION**

OAK stereo depth
GPS coords via camera pose

03



**PERSON
TRACKING**

Stationary >5s
→ flagged as still

04



**POSTURE
ANALYSIS**

BBox aspect ratio
Standing vs. fallen

05



**DASHBOARD
ALERT**

Sent only if confidence
≥ 65%



FULLY ON-DEVICE INFERENCE



LOW LATENCY



NO CLOUD REQUIRED



REAL-TIME STREAMING TO DASHBOARD

01 / YOLO DETECTION

FINE-TUNED MODEL FOR DEGRADED ENVIRONMENTS

BASE MODEL & TRAINING

Architecture **YOLO (OAK-optimized)**

Base Dataset **COCO**

Task **Person detection**

Deployment **On-device (OAK-4D)**

Alert Threshold **$\geq$ 65% confidence**

WHY FINE-TUNE?

Real emergencies involve smoke, fog, partial occlusion and poor lighting — stock COCO won't cut it.

ALBUMENTATIONS PIPELINE



RandomBrightnessContrast

Simulates over/under-exposure



RandomFog

Fog & low-visibility conditions



GaussianBlur

Motion blur, camera shake



CoarseDropout

Occlusion by debris / vegetation



RandomShadow

Uneven lighting, partial shadows



ToGray

Thermal-like grayscale input



GaussNoise

Sensor noise & image artifacts

Applied with randomized probabilities for diverse training conditions.

ON-DEVICE INTELLIGENCE

POSTURE · DEPTH · TRACKING — ALL COMPUTED ON THE CAMERA

02



DEPTH ESTIMATION

- OAK-4D stereo depth sensor

- Distance computed per detection

- GPS tracker on drone

- absolute GPS coords per person

- Demo: fixed camera coords

03



PERSON TRACKING

- Individual tracker per person

- Position sampled over time

- Stationary > 5 seconds

- flagged as STILL

- Second urgency signal

04



POSTURE ANALYSIS

- Compute BBox aspect ratio

- width > 1.3 × height

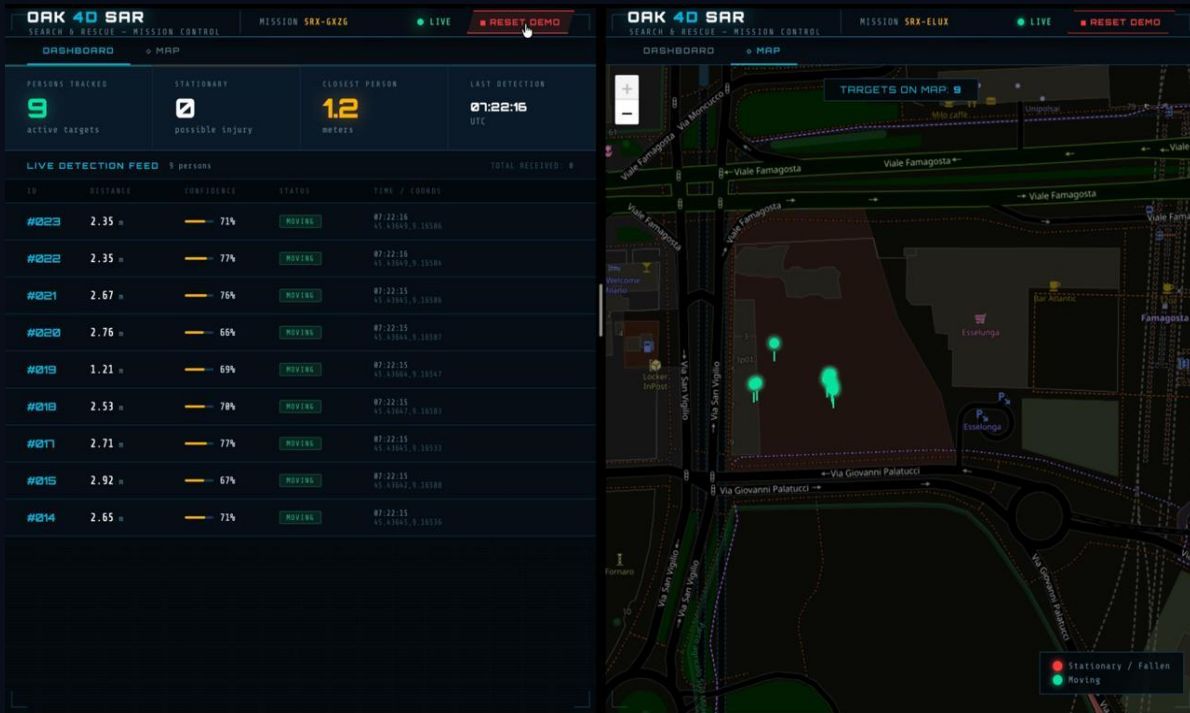
- person flagged as FALLEN

- Indicates injury or unconsciousness

- Raises intervention priority

05 / MISSION DASHBOARD

REAL-TIME OPERATOR INTERFACE



FEATURES



Live Detection Feed

All detected persons, confidence, distance



Status Indicators

MOVING (green) / STATIONARY (red)



GPS Map View

Geolocated targets on interactive map



Real-time Updates

Instant refresh per detection cycle



Mission Control

Multiple simultaneous drone streams

≥65%

CONFIDENCE GATE

5s

STILL THRESHOLD

LIVE

DATA STREAM

GPS

GEOLOCATED